

OPEN-CLOSE C1/C2/BZX/EDGX EXCHANGE 1-MINUTE INTERVAL SPECIFICATIONS

C2, BZX, EDGX (REGULAR TRADING HOURS ONLY)

File Naming Convention

{Exchange}_OpenClose_YYYY-MM-DD_HH_MM.csv.gz (Gzipped CSV)
 {Exchange} is C2, BZX, or EDGX based on product selection

Intervals

405 1-minute interval files from 09:31 – 16:15 U.S. Eastern, except on early market close days. Trade count and volume within buckets are aggregated throughout the day, not just the latest interval.

Snapshot Timing Note

The 1-minute Open-Close files are generated from real-time streaming feeds sourced directly from the Cboe exchanges' matching engines. To minimize delivery latency, files are published on a real-time cadence. During rare high-volume bursts, a small number of trades may be processed slightly late and therefore appear in the subsequent 1-minute snapshot.

C1 (GLOBAL TRADING HOURS AND REGULAR TRADING HOURS)

File Naming Convention (Effective Dec 11, 2023)

The GTH session can span multiple calendar days and have multiple time intervals for the same market date. We will continue to use the market date in the filename for all intervals but append the trading segment _1 or _2 to all files. Files in the first trading segment will have _1 in the file name and files in the second trading segment will have _2 in the file name.

Normal RTH Session

Cboe_OpenClose_YYYY-MM-DD_HH_MM_1.csv.gz (Gzipped CSV)
 Example: Market date 2024-01-29 does not span a market holiday
 Cboe_OpenClose_2024-01-29_09_31_1.csv.gz

Normal GTH Session

Cboe_OpenClose_YYYY-MM-DD_HH_MM_1.csv.gz
 Example: Market date 2024-01-29 does not span a market holiday
 Cboe_OpenClose_2024-01-29_20_16_1.csv.gz

Extended GTH Session (session spans a market holiday)

Cboe_OpenClose_YYYY-MM-DD_HH_MM_1.csv.gz
 Cboe_OpenClose_YYYY-MM-DD_HH_MM_2.csv.gz
 Example: 2024-01-15 is a Monday market holiday

Timeframe		Session	Segment	Market Date	File Naming Convention
Start	2024-01-14 20:15	GTH	1	2024-01-16	Cboe_OpenClose_YYYY-MM-DD_HH_MM_1.csv.gz
End	2024-01-15 11:30				Ex. Cboe_OpenClose_2024-01-16_20_16_1.csv.gz Ex. Cboe_OpenClose_2024-01-16_09_31_1.csv.gz
Start	2024-01-15 20:15	GTH	2	2024-01-16	Cboe_OpenClose_YYYY-MM-DD_HH_MM_2.csv.gz
End	2024-01-16 09:25				Ex. Cboe_OpenClose_2024-01-16_20_16_2.csv.gz
Start	2024-01-16 09:30	RTH	2	2024-01-16	Cboe_OpenClose_YYYY-MM-DD_HH_MM_2.csv.gz
End	2024-01-16 16:15				Ex. Cboe_OpenClose_2024-01-16_09_31_2.csv.gz Ex. Cboe_OpenClose_2024-01-16_16_15_2.csv.gz
Start	2024-01-16 16:15	Curb	2	2024-01-16	Cboe_OpenClose_YYYY-MM-DD_HH_MM_2.csv.gz
End	2024-01-16 17:00				Ex. Cboe_OpenClose_2024-01-16_16_16_2.csv.gz

Intervals

Normal Session (No market holiday)

Timeframe	Session	No. of Files	Description and File Timestamps
20:15 to 09:25	GTH	790	Aggregate volume in 1-minute intervals from 20:16 to 09:25 files

09:25 to 09:30	N/A	5	Aggregate volume in 1-minute intervals from 09:26 to 09:30 files
09:30 to 16:15	RTH	405	Aggregate volume in 1-minute intervals from 09:31 to 16:15 files
16:15 to 17:00	Curb	45	Aggregate volume in 1-minute intervals from 16:16 to 17:00 files

Note: For streaming continuity, we will continue publishing files for 9:26-9:30 (the transition between GTH and RTH), even though there is no official trading activity during this period.

File Naming Convention – Prior to Dec 11, 2023

Intervals during the GTH session on C1 are not available prior to Dec 11, 2023 and will follow the naming convention below.

Filename pattern: Cboe_OpenClose_YYYY-MM-DD_HH_MM.csv.gz (Gzipped CSV)

Trade count and volume within buckets are aggregated throughout the day, not just the latest interval.

FIELDS

#	Column Name	Data Type	Field Description
1	trade_datetime	datetime	Trade datetime (format: YYYY/MM/DD HH:MM:SS)
2	underlying	string	Underlying symbol
3	security_type	integer	Security type: 1=Equity, 2=ETF, 3=Index, 4=WCO
4	option_symbol	string	Option symbol
5	expiration_date	date	Expiration date (format YYYYMMDD)
6	strike_price	numeric	Strike price
7	call_put_flag	string	P=Put, C=Call
8	days_to_expire	integer	Number of days to expiration (calendar)
9	series_type	string	Series type. As of 2024-10-16, series type will include 'F' to indicate flex options: 'S' Standard (ex. GE, AAPL); 'N' Nonstandard (ex. GE1); 'F' Flex (ex. 2AAPL, 2QQQ1). From beginning of history to 2024-10-15, values for series type: 'S' Standard (ex. GE, AAPL); 'N' Nonstandard (2AAPL, 2QQQ1).
10	first_trade_price	numeric	Empty, not included in 1-minute intervals
11	high_trade_price	numeric	Empty, not included in 1-minute intervals
12	low_trade_price	numeric	Empty, not included in 1-minute intervals
13	last_trade_price	numeric	Empty, not included in 1-minute intervals
14	total_exchange_vol	integer	Empty, not included in 1-minute intervals
15	total_industry_vol	integer	Empty, not included in 1-minute intervals
16	open_interest	integer	Empty, not included in 1-minute intervals
17	previous_open_interest	integer	Empty, not included in 1-minute intervals
18	previous_close	numeric	Previous closing price
19	firm_open_buy_qty	integer	Firm Opening Buy - Quantity of Trades
20	firm_open_buy_vol	integer	Firm Opening Buy - Trade Volume
21	firm_close_buy_qty	integer	Firm Closing Buy - Quantity of Trades
22	firm_close_buy_vol	integer	Firm Closing Buy - Trade Volume
23	firm_open_sell_qty	integer	Firm Opening Sell - Quantity of Trades
24	firm_open_sell_vol	integer	Firm Opening Sell - Trade Volume
25	firm_close_sell_qty	integer	Firm Closing Sell - Quantity of Trades
26	firm_close_sell_vol	integer	Firm Closing Sell - Trade Volume

27	bd_open_buy_qty	integer	Broker/Dealer Opening Buy - Quantity of Trades
28	bd_open_buy_vol	integer	Broker/Dealer Opening Buy - Trade Volume
29	bd_close_buy_qty	integer	Broker/Dealer Closing Buy - Quantity of Trades
30	bd_close_buy_vol	integer	Broker/Dealer Closing Buy - Trade Volume
31	bd_open_sell_qty	integer	Broker/Dealer Opening Sell - Quantity of Trades
32	bd_open_sell_vol	integer	Broker/Dealer Opening Sell - Trade Volume
33	bd_close_sell_qty	integer	Broker/Dealer Closing Sell - Quantity of Trades
34	bd_close_sell_vol	integer	Broker/Dealer Closing Sell - Trade Volume
35	mm_buy_qty	integer	Market Maker Buy - Quantity of Trades
36	mm_buy_vol	integer	Market Maker Buy - Trade Volume
37	mm_sell_qty	integer	Market Maker Sell - Quantity of Trades
38	mm_sell_vol	integer	Market Maker Sell - Trade Volume
39	cust_lt_100_open_buy_qty	integer	Customer Buy - Quantity of Trades < 100 Contracts Each
40	cust_lt_100_open_buy_vol	integer	Customer Buy - Trade Volume < 100 Contracts Each
41	cust_lt_100_close_buy_qty	integer	Customer Buy - Quantity of Trades < 100 Contracts Each
42	cust_lt_100_close_buy_vol	integer	Customer Buy - Trade Volume < 100 Contracts Each
43	cust_lt_100_open_sell_qty	integer	Customer Sell - Quantity of Trades < 100 Contracts Each
44	cust_lt_100_open_sell_vol	integer	Customer Sell - Trade Volume < 100 Contracts Each
45	cust_lt_100_close_sell_qty	integer	Customer Sell - Quantity of Trades < 100 Contracts Each
46	cust_lt_100_close_sell_vol	integer	Customer Sell - Trade Volume < 100 Contracts Each
47	cust_100_199_open_buy_qty	integer	Customer Buy - Quantity of Trades 100-199 Contracts Each
48	cust_100_199_open_buy_vol	integer	Customer Buy - Trade Volume 100-199 Contracts Each
49	cust_100_199_close_buy_qty	integer	Customer Buy - Quantity of Trades 100-199 Contracts Each
50	cust_100_199_close_buy_vol	integer	Customer Buy - Trade Volume 100-199 Contracts Each
51	cust_100_199_open_sell_qty	integer	Customer Sell - Quantity of Trades 100-199 Contracts Each
52	cust_100_199_open_sell_vol	integer	Customer Sell - Trade Volume 100-199 Contracts Each
53	cust_100_199_close_sell_qty	integer	Customer Sell - Quantity of Trades 100-199 Contracts Each
54	cust_100_199_close_sell_vol	integer	Customer Sell - Trade Volume 100-199 Contracts Each
55	cust_gt_199_open_buy_qty	integer	Customer Buy - Quantity of Trades >=200 Contracts Each
56	cust_gt_199_open_buy_vol	integer	Customer Buy - Trade Volume >=200 Contracts Each
57	cust_gt_199_close_buy_qty	integer	Customer Buy - Quantity of Trades >=200 Contracts Each
58	cust_gt_199_close_buy_vol	integer	Customer Buy - Trade Volume >=200 Contracts Each
59	cust_gt_199_open_sell_qty	integer	Customer Sell - Quantity of Trades >=200 Contracts Each
60	cust_gt_199_open_sell_vol	integer	Customer Sell - Trade Volume >=200 Contracts Each
61	cust_gt_199_close_sell_qty	integer	Customer Sell - Quantity of Trades >=200 Contracts Each
62	cust_gt_199_close_sell_vol	integer	Customer Sell - Trade Volume >=200 Contracts Each
63	procust_lt_100_open_buy_qty	integer	Professional Customer Buy - Quantity of Trades < 100 Contracts Each
64	procust_lt_100_open_buy_vol	integer	Professional Customer Buy - Trade Volume < 100 Contracts Each
65	procust_lt_100_close_buy_qty	integer	Professional Customer Buy - Quantity of Trades < 100 Contracts Each
66	procust_lt_100_close_buy_vol	integer	Professional Customer Buy - Trade Volume < 100 Contracts Each
67	procust_lt_100_open_sell_qty	integer	Professional Customer Sell - Quantity of Trades < 100 Contracts Each

68	procust_lt_100_open_sell_vol	integer	Professional Customer Sell - Trade Volume < 100 Contracts Each
69	procust_lt_100_close_sell_qty	integer	Professional Customer Sell - Quantity of Trades < 100 Contracts Each
70	procust_lt_100_close_sell_vol	integer	Professional Customer Sell - Trade Volume < 100 Contracts Each
71	procust_100_199_open_buy_qty	integer	Professional Customer Buy - Quantity of Trades 100-199 Contracts Each
72	procust_100_199_open_buy_vol	integer	Professional Customer Buy - Trade Volume 100-199 Contracts Each
73	procust_100_199_close_buy_qty	integer	Professional Customer Buy - Quantity of Trades 100-199 Contracts Each
74	procust_100_199_close_buy_vol	integer	Professional Customer Buy - Trade Volume 100-199 Contracts Each
75	procust_100_199_open_sell_qty	integer	Professional Customer Sell - Quantity of Trades 100-199 Contracts Each
76	procust_100_199_open_sell_vol	integer	Professional Customer Sell - Trade Volume 100-199 Contracts Each
77	procust_100_199_close_sell_qty	integer	Professional Customer Sell - Quantity of Trades 100-199 Contracts Each
78	procust_100_199_close_sell_vol	integer	Professional Customer Sell - Trade Volume 100-199 Contracts Each
79	procust_gt_199_open_buy_qty	integer	Professional Customer Buy - Quantity of Trades >=200 Contracts Each
80	procust_gt_199_open_buy_vol	integer	Professional Customer Buy - Trade Volume >=200 Contracts Each
81	procust_gt_199_close_buy_qty	integer	Professional Customer Buy - Quantity of Trades >=200 Contracts Each
82	procust_gt_199_close_buy_vol	integer	Professional Customer Buy - Trade Volume >=200 Contracts Each
83	procust_gt_199_open_sell_qty	integer	Professional Customer Sell - Quantity of Trades >=200 Contracts Each
84	procust_gt_199_open_sell_vol	integer	Professional Customer Sell - Trade Volume >=200 Contracts Each
85	procust_gt_199_close_sell_qty	integer	Professional Customer Sell - Quantity of Trades >=200 Contracts Each
86	procust_gt_199_close_sell_vol	integer	Professional Customer Sell - Trade Volume >=200 Contracts Each

TRADE BUCKETING LOGIC

The volume for each option series will be bucketed by Origin using the mapping of Trade Origin/Capacity Code to Origin listed below:

Origin	Trade Origin/Capacity Code
Firm	F, L, J
Broker-Dealer	B
Market Maker	M, N
Customer	C
Pro-Customer	U, W

ORIGIN/CAPACITY CODE TYPES

Code	Capacity Type
------	---------------

B	For the account of a Broker or Dealer, including a Foreign Broker-Dealer
C	For the account of a Public Customer
F	For an OCC Clearing Member firm proprietary account
J	For a joint back-office account
L	For the account of a non-Trading Permit Holder affiliate
M	For the account of Market-Maker
N	For the account of a Market-Maker on another options exchange
U	For the account of a Professional
W	For the account of a Professional (legacy Cboe, prior to Oct 2019)